Muhammad Ahzam

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EDUCATION

Institute of Business Administration (IBA), Karachi Expected 2027
BS in Computer Science
Coursework: Software Engineering, Database Systems, Web-Based App Development, Machine Learning, Statistical Inference, Cloud Computing, NLP with Deep Learning, Data Structures, Design and Analysis of Algorithms, OOP, Operating Systems, Computer Architecture and Assembly Language, Discrete Math, Linear Algebra, Numerical Analysis

EXPERIENCE

AI-Integrated Mobile App Developer Intern (Forward Deployed), EdgeFirm Jul 15 – Aug 31, 2026

- Inherited the Weather Ways project mid-cycle and delivered its remaining three sprints to a live client, working through User Acceptance Testing and direct client feedback.
- Resolved client-reported map anomalies by rewriting coordinate-parsing logic, so traffic-flow polylines render smoothly on mobile.
- Shipped Gemini-powered, weather-aware route optimization, handling latency and state so AI output became safe, usable routes for end users.
- Built emergency-alert delivery (Twilio SMS fan-out for road closures) with global admin kill-switches as an operational safeguard.
- Added AWS CloudWatch and Firebase Crashlytics monitoring to surface silent failures after release.

Math & Problem-Solving Instructor, IBAGrads (IBA and SAT entry-test prep institute), Karachi Jul 2024 – Apr 2026

- Taught about 70 students across four batches, plus 4 private students; selected after ranking 1st in the IBA entry test.
- Tailored instruction to each batch's performance; students are now enrolled across IBA programs and abroad universities.

PROJECTS

AcademIQ: Academic Collaboration Platform (term project, 4 members; Project Lead & Full-Stack Developer) | *JavaScript, Node.js, SQL*

- Surveyed IBA students to identify needs and turned the results into requirements for the platform.
- Led a four-person team through Agile sprints, coordinating UI design with database development.
- Prioritized and built authentication, a calendar module and resource hubs.

PantryPal: Meal Planning Platform (course project, 3 members; Lead Architect & Developer) | *React, Node.js, PostgreSQL*

- Mapped user journeys and defined specifications for meal planning, inventory tracking, shopping-list PDFs, nutrition tracking and an interactive cooking mode.
- Built it end to end, connecting React UX requirements to PostgreSQL constraints (40+ API endpoints).

Hybrid RAG System for Forensic Science Queries (2-person team, equal contributors) | *Python, LangChain, Pinecone*

– Scoped a domain-specific question-answering system that answers only from a specialized PDF corpus, not general web knowledge.
– Defined faithfulness and relevancy metrics and built an evaluation framework to measure them.
– Cut response latency by 35% to 1.544 s by combining semantic and keyword retrieval.

Multi-Agent Path Finding for Hospital Pharmacy Robotics (solo) | *Python, PyTorch*

– Framed the problem from inputs and constraints (25×35 pharmacy layout, 1-tile aisles, agent priorities, time costs per move and turn) and defined cost-based metrics to benchmark routing approaches.
– Reached 100% success across six deterministic edge-case scenarios with a 1.0602 cost-efficiency ratio.

SKILLS

– **Product & Delivery:** Agile/Scrum, Sprint Planning, Requirements Definition, User Surveys, User Journey Mapping, Prioritization, Metrics Definition, Client communication.
– **Tools:** Figma, Trello, MS Excel, MS Power Apps, MS Entra, GitHub, Clickup, MS Power automate/apps
– **Technical:** SQL, PostgreSQL, React, Node.js, Python, APIs, Firebase, AWS CloudWatch

LEADERSHIP

Module Head, Enigma, IBA's flagship event 2023, 2024

– Led a 20-person team running a module for 16 competing teams (144 participants); selected two years in a row.

CERTIFICATIONS

– AI for Professionals – LUMSX/ILMx
– Business Communication – ILMx
– Supervised Learning with scikit-learn – DataCamp
– Cluster Analysis in Python – DataCamp
– Introduction to MLflow – DataCamp
– Introduction to Git – DataCamp (In Progress)